PHYSICAL AI AND SCIML IN THREE GOES · HANDBOOK

Go 3: Research Frontier: Transportation

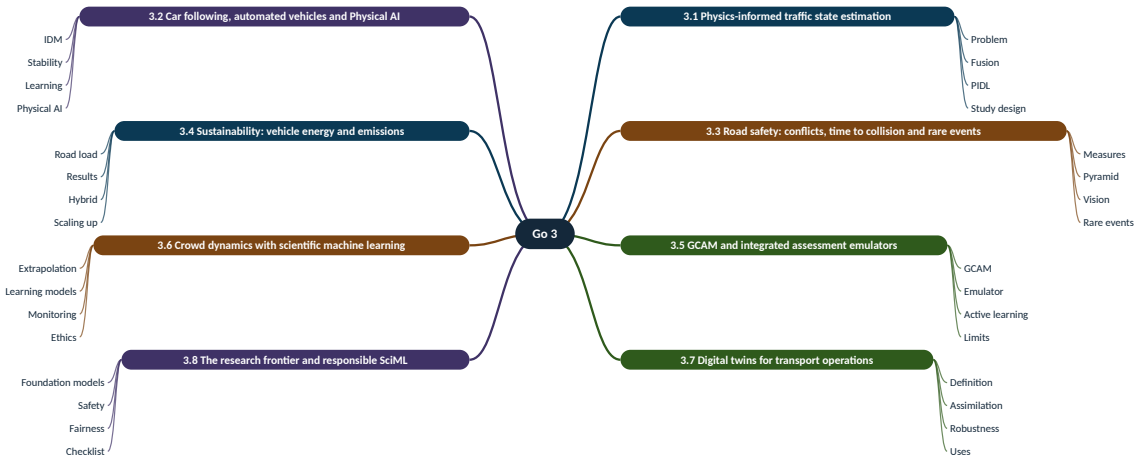
8 modules · about 6.2 hours of study

Companion to the website. Explorers, simulations, challenges and runnable code are online; this handbook keeps the text, worked numbers, mind maps and questions for reading offline.

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4. **3.4 Sustainability: vehicle energy and emissions** · 45 min
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   Integrated assessment models such as GCAM link energy, land, water, economy and climate, but each run is slow. Emulators trained on a few runs make thousands of policy scenarios and uncertainty studies possible.
6. **3.6 Crowd dynamics with scientific machine learning** · 45 min
   Crowd data are plentiful at comfortable densities and scarce at dangerous ones. Physics-shaped models, learned force parameters and physics-informed networks make crowd predictions trustworthy where it matters.
7. **3.7 Digital twins for transport operations** · 45 min
   A transport digital twin runs a model in step with live data: it estimates the current state, forecasts and tests actions safely. Data assimilation keeps it honest; physics carries it through sensor gaps.
8. **3.8 The research frontier and responsible SciML** · 40 min
   Open questions: foundation models for physical systems, rare events, verification of learned controllers and fairness. Report SciML results honestly, with baselines, uncertainty and limits.

Mind map of the go



3.1 Physics-informed traffic state estimation

50 minutes · Traffic

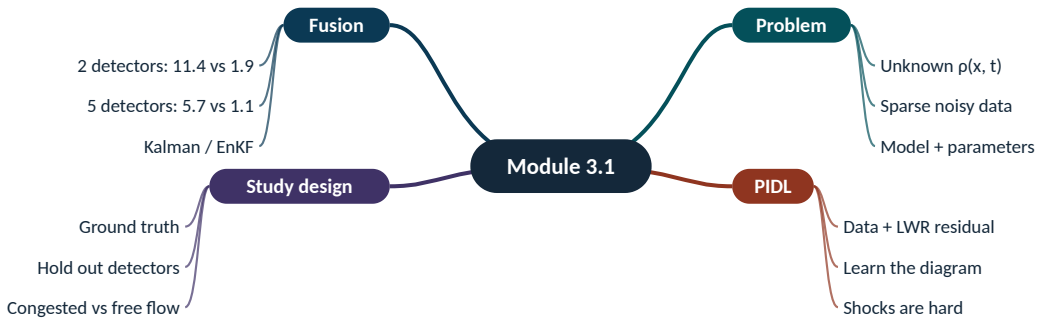
Detectors see traffic at a few points; operators need it everywhere. Combining the LWR traffic-flow model with sparse detector data reconstructs jams and shock waves that interpolation misses.

By the end you can

- state the traffic state estimation problem and its data sources
- explain why interpolation fails at shock waves
- combine the LWR model with detector data by nudging and by physics-informed deep learning
- design a study comparing estimators fairly

Overview

Traffic managers need to know density and speed along every kilometre of a road, every minute. But loop detectors sit hundreds of metres or kilometres apart, and probe vehicles give scattered samples. Filling the gaps is called traffic state estimation. The simplest approach draws straight lines between detectors, but traffic is not smooth: a jam has a sharp front, a shock wave, that moves upstream. Straight lines blur it or miss it entirely. Physics can help. The LWR model says vehicles are conserved and that flow depends on density through the fundamental diagram, so jams propagate as waves at predictable speeds. A physics-informed estimator runs this model and keeps correcting it towards the detector readings, like the Kalman filter from Go 1. Physics-informed deep learning, as in work by Shi, Mo, Di and colleagues, does the same with a neural network whose loss includes the LWR residual. In our ten kilometre ring road with two detectors, interpolation is off by about eleven vehicles per kilometre, and the physics-informed estimate by under two. In this module you will choose how many detectors to install and how much to trust them.



The estimation problem

Unknown: density ρ(x, t) along the road. Known: noisy observations at detector positions and probe locations, and a traffic model with uncertain parameters (free-flow speed, jam density, capacity). Goal: the best estimate of ρ

everywhere, with uncertainty, in real time. This is a classic inverse problem, and an ideal test case for SciML (Seo et al. 2017 review).

Model-based fusion

Run the LWR model (Godunov / cell transmission scheme, module 2.3) and at each step pull the cells near detectors towards the readings: $\rho \leftarrow \rho + g\,(z - \rho)$. In the explorer after 10 minutes on a 10 km ring road with a jam: with 2 detectors, interpolation error ≈ 11.4 veh/km versus ≈ 1.9 for the physics-informed estimate; with 5 detectors, ≈ 5.7 versus ≈ 1.1. The model carries information about the jam between detectors. The Kalman filter (and its ensemble version, EnKF) chooses the gain g optimally.

TRY Try it

Set the gain to 0: the physics model alone never learns about the jam. Raise it slowly. Is more trust in detectors always better?

Physics-informed deep learning

Physics-informed deep learning (PIDL) trains a network $\rho_\theta(x, t)$ with loss = data misfit at sensors + λ × LWR residual at collocation points, sometimes also learning the fundamental diagram with a second network (Shi et al. 2021). Strengths: flexible, can learn unknown parameters, no grid. Weaknesses: shock waves are hard for smooth networks; training can be slow for real-time use. Hybrid approaches use the network offline and a filter online.

Designing a fair study

Use a ground truth (high-resolution trajectories such as NGSIM or highD, or a microsimulation), hold out some detectors for testing, compare against interpolation and a Kalman-filter baseline, report errors in congested and free-flow conditions separately, and test robustness to detector failure and parameter error.

RESEARCH Research angle

Open question: how few detectors are needed if physics is used well? This has direct consequences for sensor investment budgets.

LAB LWR + detectors versus interpolation

`labs/go3/m31_traffic_estimation.py`

Godunov scheme for LWR, nudged towards noisy detector readings, compared with circular interpolation.

```
2 detectors: interpolation error  11.4 veh/km, physics-informed estimate  1.8 veh/km
 4 detectors: interpolation error   3.9 veh/km, physics-informed estimate  1.0 veh/km
10 detectors: interpolation error   2.9 veh/km, physics-informed estimate  1.2 veh/km
```

Takeaways

- Traffic state estimation fills gaps between sparse detectors and probes.
- Interpolation misses shock waves; the LWR model carries them between sensors.
- Model-based fusion (nudging, Kalman filters) and PIDL combine physics and data.
- Fair studies need ground truth, held-out sensors and congested-condition errors.

Check yourself

1. Why does interpolation between detectors miss jams?
   (a) Detectors are too accurate · (b) Jams have sharp moving fronts that straight lines blur · (c) Interpolation is too slow · (d) It uses physics
2. In model-based fusion, the model is…
   (a) Ignored · (b) Corrected towards detector readings at each step · (c) Replaced by data · (d) Run once at the start
3. PIDL for traffic adds to the loss…
   (a) The LWR residual · (b) A random term · (c) Only detector data · (d) Nothing
4. A fair test of an estimator should…
   (a) Use all detectors for fitting · (b) Hold out detectors and report congested-condition errors · (c) Only use free-flow data · (d) Skip baselines

Answers: 1 (b) Shock waves.; 2 (b) Predict and correct.; 3 (a) Physics term.; 4 (b) Honest evaluation.

Words

- **Traffic state estimation**: Inferring density, speed and flow everywhere on a road from sparse data.
- **Shock wave**: A sharp, moving boundary between traffic states, such as the back of a jam.
- **Nudging**: Pulling a model's state towards observations at each time step.
- **Physics-informed deep learning (PIDL)**: Training networks with physics residuals in the loss, here for traffic.
- **Ensemble Kalman filter**: A Kalman filter variant using an ensemble of model runs for nonlinear systems.

Reading

- Seo et al. (2017): Traffic state estimation on highway: a comprehensive survey: The standard review. https://doi.org/10.1016/j.arcontrol.2017.03.005
- Shi, Mo and Di (2021): Physics-informed deep learning for traffic state estimation: PIDL with a learned fundamental diagram. https://doi.org/10.1609/aaai.v35i1.16120
- highD dataset: Drone-recorded vehicle trajectories for ground truth. https://levelxdata.com/highd-dataset/

3.2 Car following, automated vehicles and Physical AI

50 minutes · Vehicles

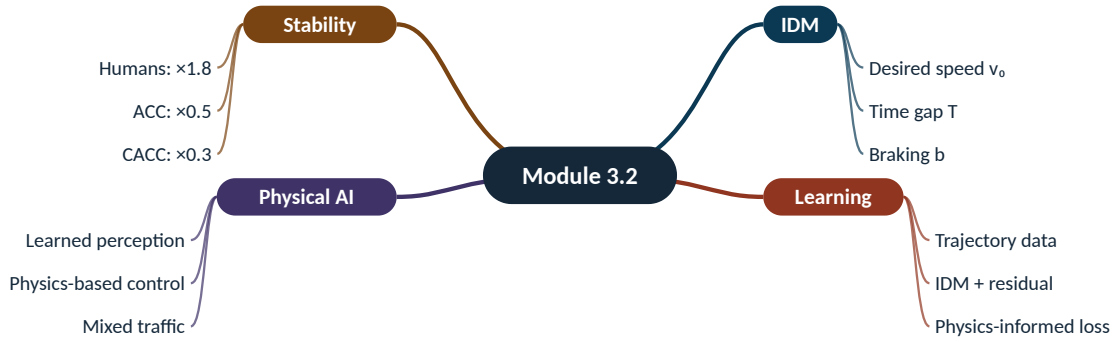
Car-following models describe how drivers react to the vehicle ahead. Physics-based models, learned residuals and cooperative automation decide whether braking waves fade or grow along a platoon.

By the end you can

- describe the Intelligent Driver Model and its parameters
- explain string stability and why human platoons amplify braking
- compare human drivers, adaptive cruise control and cooperative ACC
- outline how learned and physics models are combined in automated driving

Overview

Every driver follows the car ahead, keeping a gap that depends on speed. Car-following models write this down. The Intelligent Driver Model, by Treiber, Hennecke and Helbing, says a driver accelerates towards a desired speed but brakes as the gap shrinks below a safe distance that grows with speed and closing rate. Its few parameters, desired speed, time gap, maximum acceleration and comfortable braking, have clear physical meaning. Now put ten such drivers in a line and let the leader brake. If each follower reacts a little late and a little harder than the one ahead, the braking wave grows along the line, and the last car may stop completely. This is string instability, the origin of phantom jams. Automated vehicles are Physical AI: adaptive cruise control senses the gap by radar and reacts without delay, and cooperative adaptive cruise control also receives the leader's braking by radio, so it can damp the wave. Learned models can improve on physics, for example by adding a neural residual to the IDM, but must stay safe. In this module you will make a braking wave grow and then tame it.



The Intelligent Driver Model

IDM (Treiber, Hennecke and Helbing 2000):

$$a = a_{max}\,[1 - (v/v_0)^4 - (s^*/s)^2],\; s^* = s_0 + vT + v\Delta v \,/\, (2\sqrt{(a_{max}\,b)})$$

v_0 desired speed, T desired time gap, s_0 minimum gap, b comfortable braking, Δv closing speed. Five interpretable parameters, collision-free in normal conditions, and widely used in simulation and as a baseline for learned models.

String stability

A platoon is **string stable** if a speed dip shrinks as it travels back. In the explorer, after the leader brakes from 90 to about 40 km/h: human drivers with 0.8 s reaction amplify the dip (factor about 1.8; the last car stops), ACC with T = 1.2 s damps it (factor about 0.5), and cooperative ACC damps it further (about 0.3). Longer time gaps help stability but reduce road capacity: a real trade-off.

TRY Try it

Choose human drivers and increase the time gap from 0.6 to 2.5 s. At what gap does the wave stop growing? What does that cost in road capacity?

Learning car following

Data-driven car-following models (neural networks, Gaussian processes) learn from trajectory data such as NGSIM, highD or Waymo Open. Pure learned models can produce unsafe accelerations in rare situations. Physics-informed options: (1) a residual network on top of IDM; (2) a network predicting IDM parameters by context; (3) training with a loss that penalises IDM deviations (Mo, Shi and Di 2021). All keep collision avoidance and interpretability.

Physical AI on the road

An automated vehicle runs the sense–estimate–decide–act loop of module 1.8 at high rate: learned perception (cameras, lidar), physics-based estimation and prediction, planning and control with safety constraints. Research frontiers include learned world models that simulate traffic, verification of learned controllers, and mixed traffic where a few automated vehicles stabilise many human drivers (Stern et al. 2018).

RESEARCH Research angle

A field experiment (Stern et al. 2018) showed that a single automated vehicle could dampen stop-and-go waves among about 20 human-driven cars on a ring road.

LAB Platoons of humans, ACC and cooperative ACC

`labs/go3/m32_carfollowing.py`

IDM platoons with a human reaction delay, ACC and cooperative ACC; amplification of the braking dip.

```
human: last car's lowest speed   0.0 km/h, amplification 1.79
acc  : last car's lowest speed  65.6 km/h, amplification 0.48
cacc : last car's lowest speed  75.1 km/h, amplification 0.30
```

Takeaways

- The Intelligent Driver Model describes car following with five meaningful parameters.
- Delays and aggressive reactions make platoons string unstable, creating phantom jams.
- ACC and especially cooperative ACC can damp braking waves.
- Learned car-following models are safest when combined with physics.

Check yourself

1. String instability means…
   (a) Cars drive in a straight line · (b) Speed disturbances grow as they pass down a platoon · (c) Engines fail · (d) Gaps stay constant
2. What does cooperative ACC add to ACC?
   (a) A bigger engine · (b) Information from the vehicle ahead by radio (V2V) · (c) A human driver · (d) Longer gaps only
3. In the IDM, T is…
   (a) Top speed · (b) The desired time gap · (c) The temperature · (d) Total trip time
4. Why combine learned car-following models with physics?
   (a) To slow them down · (b) To keep behaviour safe and interpretable in rare situations · (c) It is required by law · (d) Physics is always more accurate

Answers: 1 (b) Waves grow.; 2 (b) Communication.; 3 (b) Following time gap.; 4 (b) Safety.

Words

- **Car-following model**: A model of how a driver adjusts speed in response to the vehicle ahead.
- **Intelligent Driver Model (IDM)**: A widely used car-following model with five interpretable parameters.
- **String stability**: The property that disturbances shrink as they propagate along a platoon.
- **Adaptive cruise control (ACC)**: An automated system that keeps a set gap to the vehicle ahead using radar.
- **Cooperative ACC (CACC)**: ACC that also uses wireless data from other vehicles.

Reading

- Treiber, Hennecke and Helbing (2000): Congested traffic states: The IDM paper. https://doi.org/10.1103/PhysRevE.62.1805
- Mo, Shi and Di (2021): A physics-informed deep learning paradigm for car-following models: Physics-informed car following. https://doi.org/10.1016/j.trc.2021.103240
- Stern et al. (2018): Dissipation of stop-and-go waves via control of autonomous vehicles: The ring-road field experiment. https://doi.org/10.1016/j.trc.2018.02.005

3.3 Road safety: conflicts, time to collision and rare events

50 minutes · Safety

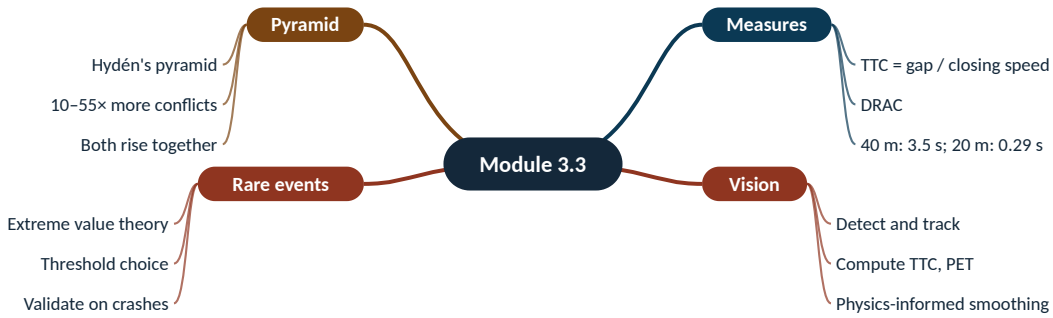
Crashes are rare, so safety studies use near misses. Physics gives surrogate measures such as time to collision; machine learning detects conflicts in video at scale; together they estimate risk before crashes happen.

By the end you can

- compute time to collision and the deceleration needed to avoid a crash
- explain why conflicts are used as surrogates for crashes
- describe how computer vision extracts conflicts from video
- outline how rare-event (extreme value) models link conflicts to crash risk

Overview

Road safety has a data problem. Crashes are thankfully rare, so waiting for enough of them to judge a new junction design takes years, and every data point is a tragedy. The alternative is to study near misses, also called traffic conflicts. Physics defines them precisely. Time to collision is the gap between two vehicles divided by their closing speed: how many seconds until impact if nobody changes course. A minimum time to collision below about one and a half seconds is a serious conflict. The deceleration rate to avoid a crash, called DRAC, measures how hard someone must brake. Machine learning makes this practical at scale: object detectors and trackers turn hours of traffic video into vehicle trajectories, and the physics measures are computed automatically. Conflicts happen far more often than crashes, and in our simulation the site with shorter gaps has both more conflicts and more crashes. Extreme value theory then links the frequency of severe conflicts to the probability of crashes. In this module you will change gap, speed, reaction and braking and watch time to collision drop.



Time to collision and DRAC

For a follower closing on a leader: **TTC = gap / ($v_{follower} - v_{leader}$)** while closing. **DRAC = (closing speed)² / (2 × gap)**, the constant deceleration needed to avoid impact. In the explorer, a 40 m gap closing at 8 m/s with a 1.5 s reaction

and 4 m/s² braking reaches a minimum TTC of 3.5 s (safe); with a 20 m gap the minimum TTC is 0.29 s and DRAC is 4 m/s² (serious conflict).

Why conflicts

The safety pyramid (Hydén 1987): many undisturbed passages, fewer slight conflicts, fewer serious conflicts, very few crashes. In the lab's simulation of 6,000 encounters, conflicts with TTC below 3 s are 10 to 55 times more frequent than crashes, and the site with shorter gaps has more of both (TTC < 3 s: 7% → 19%; crashes: 0.13% → 1.9%). Conflicts give statistical power in days instead of years.

TRY Try it

Keep the gap at 40 m and raise the reaction time from 1.5 to 3 s. When does the outcome change from safe to conflict to crash?

Machine learning at scale

Deep-learning detectors and multi-object trackers extract trajectories from video; physics computes TTC and PET on every interaction; anomaly detection flags unusual manoeuvres. Errors in tracking propagate into the measures, so trajectory smoothing and validation against manual annotation are essential. Physics-informed trajectory reconstruction (vehicles cannot teleport or accelerate at 3 g) removes tracking artefacts.

From conflicts to crash risk

Extreme value theory models the tail of conflict severity (for example, the distribution of negative TTC or of required deceleration beyond a threshold) and extrapolates to the point where a crash occurs. It requires careful threshold choice and validation against crash records; simple tails can be badly wrong. Bayesian hierarchical versions pool across sites (Zheng, Sayed and colleagues). This is an active research area.

RESEARCH Research angle

Open question: which surrogate measure (TTC, PET, DRAC or learned risk scores) best predicts crashes for vulnerable road users such as pedestrians and cyclists?

LAB Conflicts and crashes at two sites

`labs/go3/m33_safety.py`

6,000 simulated encounters per site with random gaps, speeds, reaction times and braking.

```
site A (longer gaps) : TTC < 3 s  7.32%, TTC < 1.5 s  0.52%, crashes 0.133%, conflicts (TTC <
3 s) per crash  54.9
site B (shorter gaps): TTC < 3 s 19.03%, TTC < 1.5 s  5.48%, crashes 1.850%, conflicts (TTC <
3 s) per crash  10.3
```

Takeaways

- Crashes are rare, so safety studies use conflicts measured by physics.
- TTC = gap/closing speed; DRAC = closing speed²/(2 gap).
- Computer vision extracts trajectories at scale; physics-informed smoothing removes artefacts.

- Extreme value models link conflict severity to crash risk but need careful validation.

Check yourself

1. Time to collision is…
   (a) Speed × gap · (b) Gap divided by closing speed · (c) Reaction time plus braking time · (d) Distance to the next junction
2. A minimum TTC below about 1.5 s usually indicates…
   (a) Free flow · (b) A serious conflict · (c) A parking manoeuvre · (d) Nothing
3. Why use conflicts instead of crashes?
   (a) Conflicts are more severe · (b) Conflicts are far more frequent, giving data quickly · (c) Crashes are not recorded · (d) Conflicts are easier to define legally
4. What does extreme value theory model?
   (a) The average speed · (b) The tail of conflict severity, to extrapolate to crashes · (c) Traffic volume · (d) Fuel use

Answers: 1 (b) Seconds to impact if nothing changes.; 2 (b) Common threshold.; 3 (b) Statistical power.; 4 (b) Rare events.

Words

- **Traffic conflict**: An event in which road users would collide unless one changes course or speed.
- **Time to collision (TTC)**: Time until impact if current speeds and paths continue.
- **Post-encroachment time (PET)**: Time between one road user leaving a conflict point and another arriving.
- **DRAC**: Deceleration rate needed to avoid a crash.
- **Extreme value theory**: Statistics of rare, extreme events, used to link conflicts to crashes.

Reading

- Hydén (1987): The development of a method for traffic safety evaluation (Swedish conflict technique): The safety pyramid. https://lup.lub.lu.se/search/publication/1305285
- Arun et al. (2021): A systematic review of traffic conflict-based safety measures: Review of surrogate measures and extreme value methods. https://doi.org/10.1016/j.aap.2021.106016
- Zheng, Sayed and Mannering (2021): Modeling traffic conflicts for use in road safety analysis: Methods review, including extreme value models. https://doi.org/10.1016/j.amar.2020.100142

3.4 Sustainability: vehicle energy and emissions

45 minutes · Energy

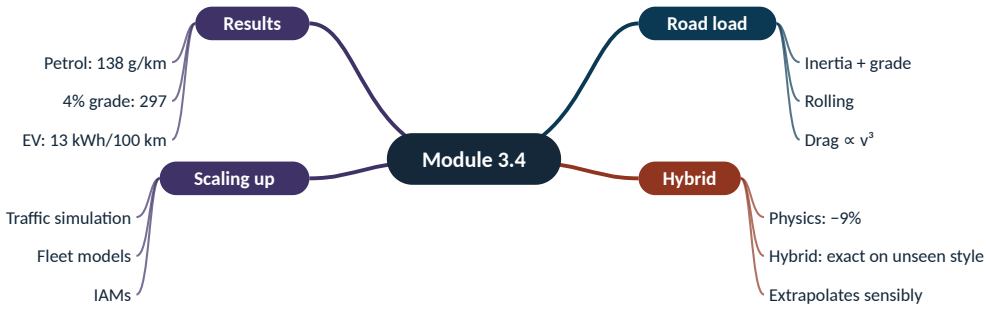
Vehicle energy follows from physics: mass, acceleration, grade, rolling resistance and drag. Hybrid models add learned corrections for engines and traffic, giving accurate, explainable emission estimates for policies.

By the end you can

- compute tractive power from the road-load equation
- explain how driving style, grade and vehicle type change energy and CO_2
- build a hybrid emission model: physics plus a learned correction
- connect vehicle-level models to network and policy analysis

Overview

Transport produces around a fifth of global carbon dioxide emissions, so estimating vehicle energy use accurately matters for every climate policy. Physics gives most of the answer. The power needed at the wheels is mass times speed times acceleration plus gravity on a grade, plus rolling resistance, plus air drag, which grows with the cube of speed. This road-load equation is behind the vehicle specific power used in the US EPA's MOVES model. A petrol engine turns only about a quarter of its fuel energy into motion, and wastes all braking energy; an electric car is about three times as efficient and recovers part of its braking energy. In our example, an aggressive driving style raises petrol CO_2 by about ten percent, a constant four percent uphill grade roughly doubles it, and the electric car uses about thirteen kilowatt hours per hundred kilometres. What physics misses, such as how engine efficiency varies with load, can be learned from measurements with a hybrid model, which keeps the physics' sound behaviour for new driving styles. In this module you will drive the same trip in different styles and vehicles.



The road-load equation

$$P = m\,v\,(a + g\sin\theta) + m\,g\,C_r\,v\cos\theta + \tfrac{1}{2}\,\rho\,C_d A\,v^3$$

For a 1,500 kg car at 20 m/s on the level: rolling ≈ 3.5 kW, drag (C_dA = 0.7 m²) ≈ 3.4 kW; accelerating at 1 m/s² adds 30 kW. Vehicle specific power (VSP, Jiménez-Palacios 1999) divides by mass and underlies the US EPA MOVES model's

operating modes.

Style, grade and vehicle

In the explorer (synthetic urban trip, about 5 km): petrol car ≈ 138 g CO_2/km; aggressive style (2× acceleration) ≈ 152 g/km; a constant 4% uphill grade ≈ 297 g/km; electric car ≈ 13 kWh/100 km, ≈ 60 g/km at 450 g/kWh grid intensity. Electric cars gain most in stop-and-go traffic because regenerative braking recovers energy that petrol cars waste.

TRY Try it

Compare petrol and electric at driving style 0.5 and 2.5. Which vehicle is more sensitive to aggressive driving, and why?

A hybrid emission model

Measured fuel use departs from the simple constant-efficiency model because efficiency varies with load. In the lab, the physics model underestimates CO_2 by about 9% (152 vs 167 g/km); fitting a small correction as a function of power on two driving styles fixes this on an unseen style (167 vs 167) and still helps for a more aggressive one (169 vs 174). The physics keeps the model sensible for new conditions; pure ML trained on one cycle would not be.

From vehicles to policy

Link vehicle models to traffic simulation (link speeds and accelerations) to estimate corridor emissions; to fleet models (EV share, vehicle age) for city scenarios; and to integrated assessment models for national and global pathways. Uncertainty in grid carbon intensity, fleet mix and driving behaviour should be carried through.

GCAM GCAM link

GCAM's transport module represents vehicle technologies by energy intensity; better vehicle-level physics and hybrid models improve those inputs.

LAB Physics vs hybrid emission model

`labs/go3/m34_energy.py`

Road-load physics with constant efficiency, plus a correction learned from 'measured' fuel on two driving styles, tested on unseen styles.

```
style x1.3: true   167 g/km, physics   152 g/km, hybrid   167 g/km
style x2.0: true   174 g/km, physics   161 g/km, hybrid   169 g/km
```

Takeaways

- Tractive power follows the road-load equation: inertia, grade, rolling and drag.
- Grade and driving style change emissions strongly; EVs recover braking energy.
- Hybrid models correct physics with measurements and extrapolate sensibly.
- Vehicle models scale to networks, fleets and integrated assessment models.

Check yourself

1. Air drag power grows with…
   (a) Speed · (b) Speed squared · (c) Speed cubed · (d) Mass
2. Why do electric cars do especially well in stop-and-go traffic?
   (a) They are lighter · (b) Regenerative braking recovers energy that petrol cars waste · (c) They have no drag · (d) They drive slower
3. In the hybrid emission model, machine learning corrects…
   (a) The road grade · (b) The physics model's error, such as varying engine efficiency · (c) The speed profile · (d) Nothing
4. Vehicle specific power is used in…
   (a) GPS receivers · (b) The US EPA MOVES emission model · (c) Traffic signals · (d) Crash tests

Answers: 1 (c) ½ ρ C_d A v^3.; 2 (b) Regen.; 3 (b) Residual learning.; 4 (b) Operating modes.

Words

- **Road-load equation**: The power needed to overcome inertia, grade, rolling resistance and aerodynamic drag.
- **Vehicle specific power (VSP)**: Tractive power per unit mass, used to classify driving for emission estimation.
- **Regenerative braking**: Recovering kinetic energy during braking to recharge a battery.
- **Grid carbon intensity**: CO_2 emitted per kWh of electricity generated.
- **MOVES**: The US EPA's MOtor Vehicle Emission Simulator.

Reading

- Jiménez-Palacios (1999): Understanding and quantifying motor vehicle emissions with vehicle specific power (MIT thesis): The origin of VSP. https://dspace.mit.edu/handle/1721.1/44505
- US EPA MOVES: The emission model and documentation. https://www.epa.gov/moves
- IEA: Global EV Outlook: Fleet and energy context for EVs. https://www.iea.org/reports/global-ev-outlook-2025

3.5 GCAM and integrated assessment emulators

50 minutes · Climate

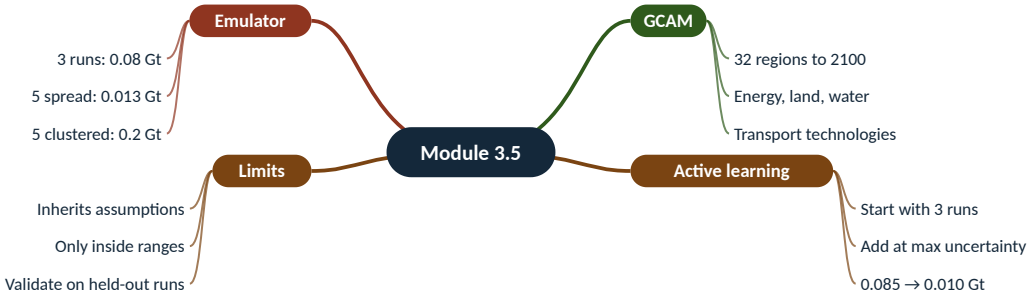
Integrated assessment models such as GCAM link energy, land, water, economy and climate, but each run is slow. Emulators trained on a few runs make thousands of policy scenarios and uncertainty studies possible.

By the end you can

- describe what GCAM represents and why its runs are expensive
- build a Gaussian process emulator of a GCAM-like transport output
- use emulator uncertainty to choose the next full-model run (active learning)
- discuss the limits of emulators for policy analysis

Overview

Integrated assessment models, or IAMs, such as GCAM, the Global Change Analysis Model developed at the Pacific Northwest National Laboratory, represent how energy, agriculture and land, water, the economy and the climate interact across the world's regions until the end of the century. They answer questions such as: what carbon price would cut transport emissions by half by 2050, and what would it do to electricity demand and land use? Each scenario can take hours to run, yet policy questions need thousands of runs to explore combinations of carbon prices, technology costs and behaviour, and to quantify uncertainty. Emulators make this feasible. We run GCAM at a carefully chosen set of inputs, train a fast statistical model, often a Gaussian process, on the outputs, and use it for everything else. The emulator's uncertainty tells us where it is unsure, so we run GCAM next exactly there: this is active learning. In our toy example, five well-placed runs already give an emulator error near one percent. In this module you will decide how many runs to buy, and where.



What GCAM represents

GCAM (Calvin et al. 2019) is an open-source integrated assessment model with 32 energy-economy regions, hundreds of land and water units, and detailed technology choices in buildings, industry and transport. Markets clear every five years; emissions feed a simple climate model. Its transport sector represents passenger and freight modes and vehicle

technologies, competing on cost through logit choice. Because everything is linked, a carbon price changes vehicle choices, electricity generation and land use at once.

Building the emulator

Pick an output, here transport CO_2 in 2050 versus carbon price. Run the full model at a few prices, fit a Gaussian process (module 2.6) and check it on extra runs. In the explorer: 3 evenly spread runs give an error of about 0.08 Gt; 5 runs about 0.013 Gt (around 1%); 5 runs bunched at low prices about 0.2 Gt, because the high-price region is unexplored.

GCAM GCAM link

Our GCAM-Learning course explains GCAM's components and how to run them. This module shows how SciML makes GCAM studies faster.

Active learning

Start with three runs, then repeatedly run the full model where the emulator's uncertainty is largest. In the lab the error falls from 0.085 Gt (3 runs) to about 0.01 Gt (7 runs), with each new run placed automatically in the least-known region. With many inputs (carbon price, battery cost, biofuel availability, behaviour), space-filling designs and dimension reduction are needed first.

TRY Try it

With 4 runs, compare evenly spread and clustered designs. Where is the uncertainty band widest in each case?

Limits and good practice

An emulator is only as good as the model it copies: it inherits GCAM's assumptions and cannot correct structural errors. It is valid only inside the sampled input ranges. Multi-output emulators must keep physical consistency (shares summing to one, emissions consistent with energy). Report emulator uncertainty alongside model uncertainty, and validate on held-out full runs. Published GCAM emulators and 'fast IAM' surrogates are an active research area.

LAB A GP emulator with active learning

`labs/go3/m35_emulator.py`

Start with three runs; add each new run where the emulator is most uncertain.

```
3 runs: emulator error 0.0845 Gt, largest uncertainty at 300 $/t
4 runs: emulator error 0.0768 Gt, largest uncertainty at 95 $/t
5 runs: emulator error 0.0125 Gt, largest uncertainty at 354 $/t
6 runs: emulator error 0.0126 Gt, largest uncertainty at 43 $/t
7 runs: emulator error 0.0104 Gt, largest uncertainty at 152 $/t
```

Takeaways

- GCAM links energy, land, water, economy and climate across regions, but runs are slow.
- A Gaussian process emulator trained on a few runs answers thousands of policy questions.
- Active learning places new runs where the emulator is least certain.

- Emulators inherit the model's assumptions and are valid only inside sampled ranges.

Check yourself

1. GCAM is…
   (a) A traffic simulator · (b) An integrated assessment model linking energy, land, water, economy and climate · (c) A crash database · (d) A GPS service
2. Why emulate GCAM?
   (a) It is inaccurate · (b) Each run is slow, and policy studies need many runs · (c) It has no outputs · (d) Emulators replace data
3. Active learning chooses the next full run…
   (a) At random · (b) Where the emulator is most uncertain · (c) At zero carbon price · (d) Where it is most certain
4. An emulator can correct GCAM's structural errors.
   (a) True · (b) False, it inherits them · (c) Only with more runs · (d) Only for transport

Answers: 1 (b) Global Change Analysis Model.; 2 (b) Speed for many scenarios.; 3 (b) Learn where unsure.; 4 (b) A copy, not the truth.

Words

- **Integrated assessment model (IAM)**: A model linking energy, economy, land use and climate to evaluate policies.
- **GCAM**: The Global Change Analysis Model, an open-source IAM from PNNL's Joint Global Change Research Institute.
- **Emulator**: A fast statistical approximation of a slow simulation model.
- **Active learning**: Choosing new model runs where an emulator is most uncertain.
- **Carbon price**: A charge per tonne of CO_2 emitted, used in scenarios to drive emission reductions.

Reading

- Calvin et al. (2019): GCAM v5.1: The model description paper. https://doi.org/10.5194/gmd-12-677-2019
- GCAM documentation: Official documentation. https://jgcri.github.io/gcam-doc/
- GCAM-Learning: Our companion course on GCAM in three goes. https://smrahmanbd.github.io/GCAM-Learning/

3.6 Crowd dynamics with scientific machine learning

45 minutes · Crowds

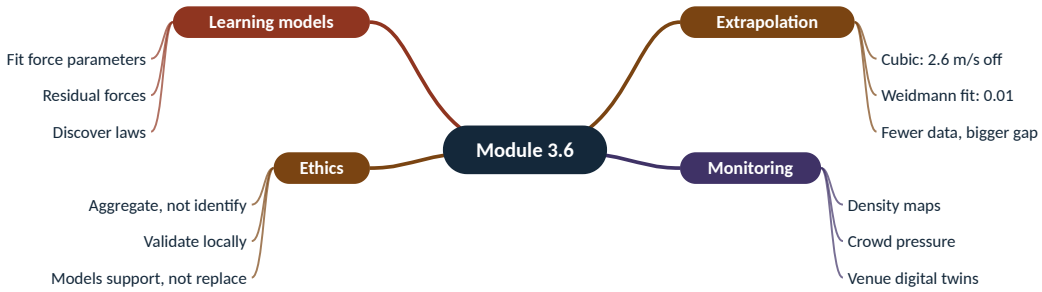
Crowd data are plentiful at comfortable densities and scarce at dangerous ones. Physics-shaped models, learned force parameters and physics-informed networks make crowd predictions trustworthy where it matters.

By the end you can

- explain why crowd models must extrapolate to dangerous densities
- fit a physics-shaped fundamental diagram and compare it with a free curve
- describe how SciML learns social-force or route-choice parameters from trajectories
- outline physics-informed crowd monitoring and its ethics

Overview

At mass gatherings, from stadiums to the Hajj, crowd safety depends on predicting what happens at high densities. Yet most measurements come from comfortable conditions, because dangerous densities are rare and hard to measure. This is exactly the setting where pure data fits fail. Suppose we measure walking speed at densities up to two and a half persons per square metre and fit a flexible cubic curve. At five persons per square metre, the cubic predicts people walking faster than at low density, which is physically absurd. A physics-shaped law, such as Weidmann's fundamental diagram, where speed falls to zero at a jam density, needs only its three parameters fitted, and extrapolates almost perfectly. The same idea scales up: learning the parameters of social-force or other pedestrian models from trajectory data, physics-informed networks for crowd density fields, and hybrid route-choice models. Physical AI enters too, through camera-based crowd counting and real-time digital twins of event spaces, which raise privacy questions. In this module you will see a free curve fail where a physics-shaped one succeeds.



Extrapolating to danger

Weidmann (1993): $v(\rho) = v_0\,(1 - \exp(-\gamma\,(1/\rho - 1/\rho_{max})))$, with $v_0 = 1.34$ m/s, $\gamma = 1.913$, $\rho_{max} = 5.4$ persons/m². In the explorer, with data only up to 2.5 persons/m², a cubic fit is off by about 2.6 m/s beyond the data (it predicts people

speeding up at high density), while fitting only Weidmann's three parameters is off by about 0.01 m/s. With data only up to 1.5 persons/m², the cubic error exceeds 15 m/s; the physics-shaped fit stays near 0.015.

Learning pedestrian models

Social-force and similar models have parameters (desired speed, interaction strength and range, relaxation time) that vary by culture, purpose and setting. SciML calibrates them from trajectories by minimising simulated-versus-observed differences, or learns residual interaction forces with neural networks while keeping the physical structure (Kreiss et al. on social LSTMs; physics-informed hybrids). Equation discovery (module 2.4) has been used to find pedestrian interaction laws from data.

TRY Try it

Lower the highest observed density to 1.5 persons/m². What does the cubic predict at 5 persons/m²? Does the physics-shaped fit change much?

Monitoring with Physical AI

Crowd-counting networks estimate density maps from camera images; optical flow gives velocities; crowd pressure (density × velocity variance, Helbing et al. 2007) flags turbulence. Physics-informed post-processing enforces conservation between camera views. Digital twins of venues combine these with simulation to forecast density minutes ahead. Our Crowd Dynamics course covers the crowd science in depth.

RESEARCH Research angle

See our Crowd Dynamics in three goes course for mass-gathering crowd science; SciML adds learned parameters and physics-informed monitoring on top.

Ethics

Crowd monitoring must protect privacy: use aggregate density and flow rather than identities, process video on site, minimise retention, and be transparent about purpose. Models trained on one culture or venue may not transfer; validate locally, and never let a model replace trained stewards and clear authority to act.

LAB Physics-shaped versus free fit for crowds

`labs/go3/m36_crowd.py`

Grid-search Weidmann's three parameters versus fitting a cubic, on light-crowd data only.

```
fitted v0 = 1.40 m/s, gamma = 1.65, rho_max = 5.7 per m2
error at 2.6-5.4 per m2: cubic 1.66 m/s, physics-shaped 0.012 m/s
cubic predicts 2.71 m/s at 5 per m2 (true 0.04)
```

Takeaways

- Crowd data are plentiful at safe densities and scarce at dangerous ones.
- Physics-shaped laws such as Weidmann's extrapolate safely; free curves do not.
- SciML calibrates and corrects pedestrian models from trajectory data.
- Camera-based monitoring needs physics checks and strong privacy safeguards.

Check yourself

1. Why must crowd models extrapolate?
   (a) Data at dangerous densities are scarce · (b) Crowds never change · (c) Cameras are perfect · (d) Physics is unknown
2. At high density, an unconstrained cubic fitted to light-crowd data may predict…
   (a) Zero speed · (b) Physically absurd speeds, even speeding up · (c) The exact truth · (d) No output
3. Fitting Weidmann's law means estimating…
   (a) Hundreds of weights · (b) Three parameters: v_0, γ and ρ_{max} · (c) Only the mean speed · (d) Camera angles
4. A privacy-respecting crowd monitor should…
   (a) Identify every person · (b) Use aggregate densities and minimise retention · (c) Share video publicly · (d) Avoid validation

Answers: 1 (a) Danger is where data are missing.; 2 (b) Free curves wander.; 3 (b) Few physical parameters.; 4 (b) Aggregate, not identify.

Words

- **Fundamental diagram (crowds)**: The relation between pedestrian density, speed and flow.
- **Weidmann's law**: A widely used pedestrian speed–density relation with three parameters.
- **Social force model**: A pedestrian model in which people respond to attractive and repulsive forces.
- **Crowd counting**: Estimating the number or density of people from images.
- **Crowd pressure**: Density times velocity variance; an indicator of crowd turbulence.

Reading

- Weidmann (1993): Transporttechnik der Fussgänger: Source of the pedestrian fundamental diagram. https://doi.org/10.3929/ethz-a-000687810
- Helbing, Johansson and Al-Abideen (2007): Dynamics of crowd disasters: Crowd pressure and turbulence at the Hajj. https://doi.org/10.1103/PhysRevE.75.046109
- Crowd Dynamics in three goes: Our companion course on crowd science. https://smrahmanbd.github.io/Crowd-Dynamics-Learning/

3.7 Digital twins for transport operations

45 minutes · Twins

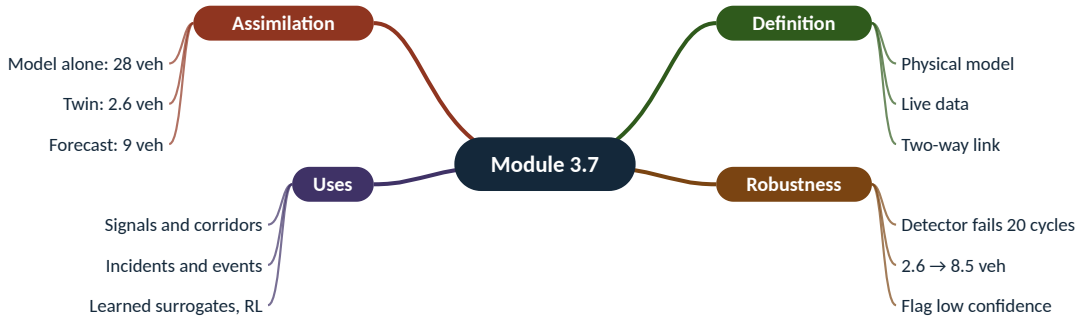
A transport digital twin runs a model in step with live data: it estimates the current state, forecasts and tests actions safely. Data assimilation keeps it honest; physics carries it through sensor gaps.

By the end you can

- define a transport digital twin and its components
- explain data assimilation as predict-and-correct at system scale
- show how a twin survives a sensor failure
- describe uses for signals, motorways and resilience planning

Overview

A digital twin is more than a simulation. It is a model of a real road, junction or city that is continuously corrected with live data, so that at any moment it represents the current state of the real system. Operators can then look into the near future: forecast queues in the next fifteen minutes, test a new signal plan or a diversion in the twin before applying it, and plan for emergencies. The core mechanism is data assimilation, the same predict-and-correct idea as the Kalman filter, applied to a whole traffic model. Our example is a signalised approach. The queue model alone has an arrival estimate that is fifteen percent too low, so it drifts far from reality, with an error of about twenty-eight vehicles. Corrected each cycle with a noisy detector, the twin tracks the real queue within about three vehicles. When the detector fails for twenty cycles, the model carries the twin through the gap and the error grows only modestly. Twins combine physics models, machine learning for forecasting and perception, and Physical AI for control. In this module you will tune a twin and break its detector.



What makes a twin

A digital twin needs (1) a model of the physical system, (2) live data, (3) a two-way link: data update the model, and the model informs actions on the system. Without live updating it is a simulation; without a link to decisions it is a

dashboard. Transport twins range from one junction to whole cities (for example, digital twins built on traffic simulators such as SUMO or Aimsun with real-time feeds).

Data assimilation in practice

Each cycle: predict the queue with the model, then correct towards the detector by a gain g. In the explorer: model alone, error ≈ 28 vehicles (it underestimates arrivals by 15%); twin with g = 0.5, ≈ 2.6 vehicles; three-cycle-ahead forecasts ≈ 9 vehicles. Too low a gain ignores data; too high follows detector noise. For large networks, ensemble Kalman filters and particle filters are used.

TRY Try it

Set the gain to 0 (model only), then 1 (trust the detector fully) with noise 10. Which gives the best twin?

Surviving sensor failures

When the detector fails for 20 cycles, the twin keeps running the model; the error grows from about 2.6 to about 8.5 vehicles, still far better than the uncorrected model. Physics-based structure is what makes this graceful degradation possible; a pure data model would have nothing to run on. Twins should flag reduced confidence during gaps.

Uses and frontiers

Signal timing and corridor management; incident response and diversion testing; event and evacuation planning; resilience studies (floods, heat). Frontiers: learned surrogates (module 2.5) to make city-scale twins real-time, Physical AI controllers trained in twins (reinforcement learning) and transferred safely to the real world, and standards for twin validation.

RESEARCH Research angle

Open question: how should a city decide that a digital twin is good enough to test interventions? Validation standards for transport twins are still emerging.

LAB A queue digital twin with a detector failure

`labs/go3/m37_twin.py`

Model alone, twin with assimilation, and twin through a 20-cycle detector failure.

```
gain 0.0, detector working: model alone  41.9 veh, twin  41.9 veh
gain 0.5, detector working: model alone  41.9 veh, twin   3.6 veh
gain 0.5, detector fails 20 cycles: model alone  41.9 veh, twin   7.8 veh
```

Takeaways

- A digital twin is a model kept in step with a real system by live data and linked to decisions.
- Data assimilation predicts with physics and corrects with measurements.
- Physics structure lets a twin degrade gracefully when sensors fail.
- Twins support signals, incidents, events and resilience; learned surrogates make them faster.

Check yourself

1. What distinguishes a digital twin from a simulation?
   (a) Better graphics · (b) Continuous updating with live data and a link to decisions · (c) More vehicles · (d) Nothing
2. Data assimilation means…
   (a) Deleting old data · (b) Correcting a model's predictions with observations · (c) Collecting more sensors · (d) Training once
3. When a detector fails, a physics-based twin…
   (a) Stops · (b) Keeps running its model with growing uncertainty · (c) Becomes perfect · (d) Uses random numbers
4. A twin can test a new signal plan…
   (a) Only after applying it · (b) Safely before applying it to the real junction · (c) Never · (d) Only at night

Answers: 1 (b) Two-way link.; 2 (b) Predict and correct.; 3 (b) Graceful degradation.; 4 (b) What-if analysis.

Words

- **Digital twin**: A model continuously updated with data from, and linked to decisions about, a real system.
- **Data assimilation**: Combining model predictions with observations to estimate a system's state.
- **Graceful degradation**: Continuing to work, with reduced accuracy, when parts fail.
- **Particle filter**: A data assimilation method using many weighted model samples.
- **What-if analysis**: Testing possible actions in a model before applying them.

Reading

- Kušić et al. (2023): A digital twin in transportation: real-time synergy of traffic data streams and simulation: A motorway digital twin. https://doi.org/10.1016/j.aei.2022.101858
- SUMO: Simulation of Urban Mobility: Open-source traffic simulator used in many twins. https://eclipse.dev/sumo/
- Transport Twin Study Path: Our companion course on transport digital twins. https://smrahmanbd.github.io/transport-twin-study/

3.8 The research frontier and responsible SciML

40 minutes · Research

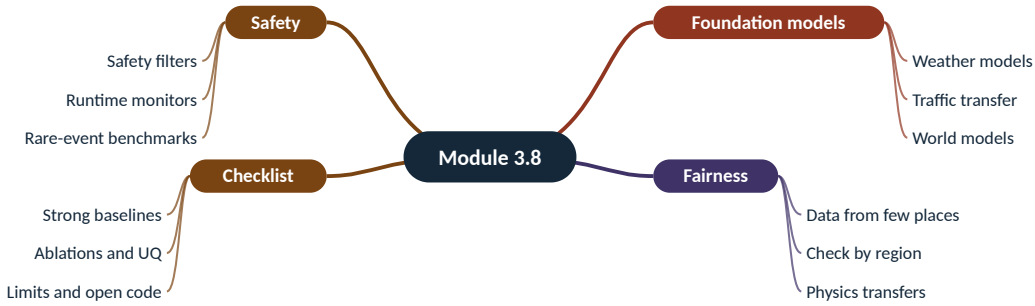
Open questions: foundation models for physical systems, rare events, verification of learned controllers and fairness. Report SciML results honestly, with baselines, uncertainty and limits.

By the end you can

- name open research questions in Physical AI and SciML for transport
- describe foundation models and world models for physical systems
- apply a checklist for honest SciML reporting
- discuss ethical and fairness issues in transport SciML

Overview

You have now met the main tools of scientific machine learning and seen them at work in traffic, vehicles, safety, energy, climate models, crowds and digital twins. Many questions remain open. Can large foundation models, trained on huge collections of simulations and sensor data, learn general physical behaviour that transfers to new cities and new problems? Can learned world models simulate traffic well enough to train automated vehicles safely? How do we verify that a learned controller will never do something dangerous in a situation it has never seen? How do we model rare events, such as crashes, crushes and extreme weather, when data on them are almost absent? Who benefits and who is left out when models trained in one place are used in another? Answering these needs good research practice. Always compare against strong classical baselines and simple models. Report uncertainty, not just a single error. Test on conditions the model has not seen. State the limits clearly, and share code and data. In this module you will turn these ideas into a checklist for your own project.



Foundation and world models

Foundation models pretrained on huge amounts of physical data (for example, weather models such as GraphCast and Aurora) now rival traditional forecasting systems. In transport, open questions include pretrained traffic models that

transfer between cities, and learned world models that generate realistic traffic scenes for training automated vehicles. Do they respect physics, and how do we know when they are wrong?

Safety, rare events and verification

Learned controllers must be verified: reachability analysis, safety filters (control barrier functions) and runtime monitors that fall back to physics-based control. Rare events need simulation, physics constraints and extreme value methods, because data barely exist. Benchmarks with honest rare-event test sets are scarce and valuable.

RESEARCH Pick your front

Choose one front for your project and say which open question you address, which data and physics you use, and how you will know if you are wrong.

Fairness and transfer

Most training data come from a few well-instrumented countries and cities. Models may perform worse in places with different vehicles, road users, cultures or climates, including many cities in the Global South. Check performance by region and group, involve local stakeholders, and prefer physics-based structure that transfers better than pure correlations.

A checklist for honest SciML

Before submitting: (1) a precise question and metric; (2) leakage-free splits; (3) strong classical and simple baselines; (4) physics consistency checks; (5) uncertainty with calibration evidence; (6) out-of-distribution tests; (7) ablations showing what the physics adds; (8) computational cost compared fairly; (9) limits and failure cases stated; (10) code, data and seeds shared. McGreivy and Hakim (2024) found weak baselines common in ML-for-PDE papers: do better.

TRAP Common trap

Claiming a PINN 'beats' a solver after tuning the PINN for weeks and the solver for minutes. Compare at equal effort, accuracy or cost.

LAB Report a result with a baseline and an interval

Compare paired errors against a baseline and report a bootstrap interval, not a single number.

```
baseline 5.6, SciML 1.3 veh/km
improvement 4.3 veh/km, 95% bootstrap interval 3.9 to 4.7; better on 100% of days
```

Takeaways

- Open fronts: foundation and world models, verification, rare events, fairness and transfer.
- Learned controllers need safety filters, monitors and physics fallbacks.
- Check performance across regions and groups; physics structure transfers better.
- Use the ten-point checklist: baselines, UQ, ablations, limits and open code.

Check yourself

1. A learned world model is…
   (a) A globe · (b) A generative model of scenes used to train and test systems · (c) A map app · (d) A survey
2. A safety filter for a learned controller…
   (a) Removes the controller · (b) Overrides actions that would break safety constraints · (c) Adds noise · (d) Speeds it up
3. Why check model performance by region?
   (a) It is required by PyTorch · (b) Models trained in data-rich places may fail elsewhere · (c) Regions have equal data · (d) It improves training speed
4. An ablation study shows…
   (a) Model size · (b) What each component, such as the physics term, contributes · (c) Training time · (d) Data volume

Answers: 1 (b) Learned simulator.; 2 (b) Barrier functions.; 3 (b) Fairness and transfer.; 4 (b) Remove parts and compare.

Words

- **Foundation model**: A large model pretrained on broad data and adapted to many tasks.
- **World model**: A learned model that simulates how an environment evolves.
- **Safety filter**: A mechanism that modifies a controller's actions to keep the system safe.
- **Ablation study**: Removing parts of a model to measure their contribution.
- **Out-of-distribution test**: Evaluation on conditions unlike the training data.

Reading

- Lam et al. (2023): GraphCast: A machine-learning weather model rivalling traditional forecasts. https://doi.org/10.1126/science.adi2336
- McGreivy and Hakim (2024): Weak baselines and reporting biases in ML for fluid-related PDEs: Why strong baselines matter. https://doi.org/10.1038/s42256-024-00897-5
- Di and Shi (2021): A survey on autonomous vehicle control in the era of mixed-autonomy: Physical AI on the road. https://doi.org/10.1016/j.trc.2021.103008

Questions for spaced practice

From the question bank: answer from memory, then check. Revisit after 1, 3, 7 and 14 days.

3.1 Physics-informed traffic state estimation

1. **Recall:** What is traffic state estimation?
   → Estimating density, speed and flow along a whole road over time from sparse, noisy measurements.
2. **Explain:** Why does physics help between detectors?
   → The LWR model propagates jams as waves at predictable speeds, so information moves correctly between sensors.
3. **Explain:** What does the nudging gain control?
   → How strongly the model is pulled towards detector readings: too low ignores data, too high follows noise.
4. **Recall:** Name two sources of traffic data.
   → Loop detectors, probe vehicles (GPS, connected vehicles), cameras, Bluetooth readers.
5. **Apply:** Design a test for a new estimator on a motorway.
   → Use trajectory ground truth or simulation, hold out detectors, compare with interpolation and a Kalman baseline, report congested errors separately.
6. **Explain:** Why are shock waves hard for PINN-type estimators?
   → Smooth networks struggle to represent sharp discontinuities (spectral bias), so fronts get smeared.

3.2 Car following, automated vehicles and Physical AI

1. **Recall:** Name the IDM parameters.
   → Desired speed v0, time gap T, minimum gap s0, maximum acceleration a, comfortable braking b.
2. **Explain:** Why do human platoons amplify braking?
   → Reaction delays and over-reaction mean each follower brakes later and harder than the vehicle ahead.
3. **Explain:** How does CACC improve string stability?
   → It receives the leader's acceleration by radio and reacts immediately, before the gap changes.
4. **Apply:** What trade-off does a longer time gap bring?
   → Better stability and safety but lower road capacity (fewer vehicles per hour).
5. **Recall:** Name a public trajectory dataset for car following.
   → NGSIM, highD, Waymo Open, Lyft Level 5.
6. **Explain:** How can a neural network refine the IDM safely?
   → By learning a residual on top of IDM acceleration or predicting IDM parameters, so the physics still prevents collisions.

3.3 Road safety: conflicts, time to collision and rare events

1. **Calculate:** Gap 30 m, follower 20 m/s, leader 15 m/s. TTC?
   → 30 / (20 − 15) = 6 s.
2. **Calculate:** DRAC for 8 m/s closing speed and 16 m gap?
   → $8^2 / (2 \times 16) = 2$ m/s².
3. **Explain:** Why are conflicts useful surrogates for crashes?
   → They are much more frequent and physically related to crashes, so risk can be assessed in days rather than years.
4. **Explain:** How does machine learning help conflict studies?
   → Detectors and trackers extract trajectories from video automatically, allowing TTC and PET to be computed for every interaction.
5. **Apply:** How would you remove tracking artefacts from trajectories?
   → Physics-informed smoothing: constrain speeds and accelerations to physically possible values, then validate against manual annotations.

6. **Explain:** Why must extreme value models be validated on crash data?
   → Tail extrapolation is sensitive to thresholds and assumptions and can be badly wrong.

3.4 Sustainability: vehicle energy and emissions

1. **Recall:** Write the road-load equation.
   → $P = m\,v\,(a + g\sin\theta) + m\,g\,C_r\,v\cos\theta + \tfrac{1}{2}\,\rho\,C_d\,A\,v^3$.
2. **Calculate:** Drag power at 20 m/s with $\rho = 1.2$, $C_dA = 0.7$?
   → $0.5 \times 1.2 \times 0.7 \times 20^3 = 3{,}360$ W ≈ 3.4 kW.
3. **Explain:** Why does an uphill grade raise emissions so much?
   → Lifting the car's weight adds $m\,g\sin\theta\,v$ to the power, comparable to or larger than rolling and drag at urban speeds.
4. **Explain:** What does the learned part of a hybrid emission model capture?
   → Effects the simple physics omits: load-dependent engine efficiency, cold starts, auxiliary loads.
5. **Apply:** How would you estimate corridor CO_2 for a new speed limit?
   → Simulate traffic to get speed and acceleration profiles, apply the hybrid vehicle model per vehicle type, and sum with fleet shares and uncertainty.
6. **Explain:** Why does an EV's CO_2 per km depend on location?
   → Its emissions come from electricity generation, whose carbon intensity varies by grid and time.

3.5 GCAM and integrated assessment emulators

1. **Recall:** What does GCAM couple?
   → Energy, agriculture and land, water, the economy and the climate, across 32 regions.
2. **Explain:** Why are emulators valuable for IAM policy studies?
   → Each full run is slow; emulators answer thousands of scenario and uncertainty queries in seconds.
3. **Explain:** What is active learning in emulation?
   → Iteratively choosing the next full-model run where the emulator's uncertainty is largest.
4. **Apply:** Your emulator is trained on carbon prices 0–200 $/t. Can you use it at 350 $/t?
   → No: it is only valid within the sampled range; run GCAM there first.
5. **Explain:** Name two limitations of emulators.
   → They inherit the model's structural assumptions and are only valid within sampled input ranges; multi-output consistency must be checked.
6. **Apply:** Which inputs might a GCAM transport emulator vary?
   → Carbon price, battery costs, fuel prices, biofuel availability, travel demand elasticities.

3.6 Crowd dynamics with scientific machine learning

1. **Recall:** Write Weidmann's speed–density law.
   → $v = v_0\,(1 - \exp(-\gamma\,(1/\rho - 1/\rho_{max})))$, with $v_0 = 1.34$ m/s, $\gamma = 1.913$, $\rho_{max} = 5.4$ per m².
2. **Explain:** Why does a free cubic fail at high density?
   → It is fitted only to low-density data and has no constraint forcing speed to fall to zero, so it can curve upwards.
3. **Explain:** How can SciML improve social-force models?
   → By calibrating their parameters from trajectories and learning residual forces with networks while keeping the physical structure.
4. **Apply:** Design a physics check for camera density maps.
   → Check that people are conserved between overlapping views and over time, given measured flows through boundaries.
5. **Explain:** Name two privacy safeguards for crowd monitoring.
   → Aggregate densities instead of identities, on-site processing, minimal retention, transparency about purpose.
6. **Apply:** Why validate a crowd model locally before a pilgrimage?
   → Walking speeds and behaviour vary with culture, age, purpose and setting, so imported parameters may mislead.

3.7 Digital twins for transport operations

1. **Recall:** What three components does a digital twin need?
   → A model of the system, live data, and a two-way link to decisions.
2. **Explain:** How does data assimilation keep a twin accurate?
   → Each step, it predicts with the model and corrects the state towards new observations by a gain based on their uncertainties.
3. **Explain:** Why can a physics-based twin survive sensor gaps?
   → Its model keeps predicting how the system evolves, so the estimate degrades gradually instead of disappearing.
4. **Apply:** Name two uses of a corridor digital twin.
   → Forecasting congestion, testing signal plans or diversions, incident response, event planning.
5. **Explain:** What happens with too high an assimilation gain and noisy detectors?
   → The twin follows detector noise, making its state and forecasts jumpy.
6. **Apply:** How could learned surrogates help a city-scale twin?
   → They replace slow simulators with fast operator-learning or emulator models so forecasts and what-ifs run in real time.

3.8 The research frontier and responsible SciML

1. **Recall:** Name three open research fronts in SciML for transport.
   → Foundation and world models, verification of learned control, rare events, fairness and transfer.
2. **Explain:** Why are rare events hard for machine learning?
   → There are almost no data on them, so models must rely on physics, simulation and careful extrapolation.
3. **Explain:** What does a runtime safety monitor do?
   → Checks a learned controller's actions online and switches to a safe fallback when limits might be violated.
4. **Apply:** List five items from the honest-reporting checklist.
   → Leakage-free splits, strong baselines, uncertainty with calibration, out-of-distribution tests, ablations, limits, open code.
5. **Explain:** Why might a traffic model trained in one country fail in another?
   → Different vehicles, road users, behaviour, infrastructure and climate shift the data distribution.
6. **Apply:** How would you show what physics adds to your model?
   → Run an ablation: train the same model without the physics term and compare accuracy and robustness on held-out and out-of-distribution data.